CP3406 Assessment 1: Utility App

Overview

Utility apps provide rapid, "at-a-glance" information to the user. For example, a simplified local weather app shows only the most relevant information about the local temperature and current weather conditions. Another example is a simple mindfulness app that helps the user keep track of how long they meditate.

In this task, you will design and build a utility-style mobile application that delivers focused, at-a-glance functionality for a specific daily-life activity. You are free to choose which activity, what aspect of daily life, and the design and functionality of the utility app. User-interaction via button events is expected to be minimal but intuitive. Your utility app must consist of a *main screen* and a *settings screen* that controls some aspects of the main screen content or functionality. Note that for this app, the settings do not need to be persistent.

Additionally, you will write a brief *self-reflection* about meaningful experiences that occurred while working on your utility app.

Technical Scope

At a minimum, your utility app should demonstrate your understanding of the topic areas covered in Weeks 1-5 (shown in the table). However, we may encourage you to explore additional aspects of app development to enhance your utility app in interesting and useful ways.

Week	Topic
1	Kotlin and Android Studio setup
2	Layouts using Jetpack Compose
3	Material Design 3 principles
4	App architecture (ViewModel, DI, Repository pattern)
5	Web APIs using Retrofit

Starter Template

You will begin with a starter template, which includes several Composables:

- UtilityApp() – the main scaffold with bottom navigation
- UtilityScreen() – placeholder for your core utility functionality
- SettingsScreen() – placeholder for user preferences or configuration

The template uses Jetpack Compose and basic Material Design 3, and toggles between screens using a bottom navigation bar. This avoids complex navigation (not covered until Week 6) and keeps everything within a single activity.

Detailed Requirements

1. **App Development**
    - Your app should be developed using Kotlin and modern Android APIs, libraries, and coding principles.
    - Follow Material Design principles with Jetpack Compose for creating a clean, responsive, and user-friendly interface.
    - Use ViewModels, Repository pattern, dependency injection and other architecture best practices to manage app data and business logic and support good Android lifecycle management.
    - Simple navigation is already implemented in the starter template. You are not expected to have additional screens or navigation beyond the provided code.
    - Implement networking capabilities to fetch data from relevant external APIs or services to provide novel and creative features in your app.
    - Use GitHub for version control to manage and document your code and commit regularly (at least a few times each week) to clearly show your development progress. In addition, provide a README file that clearly outlines the core features of your app and their implementation.
2. **Self-Reflection**
    - You will need to submit a 1000-word self-reflection based on **Gibbs' Reflective Cycle**. The purpose of this self-reflection is to critically assess your work, identify challenges, and highlight areas of improvement.
    - Focus on:
        - *Description:* What happened during the project?
        - *Feelings:* What were your feelings/reactions throughout the project?
        - *Evaluation:* What went well and what didn't go well?
        - *Analysis:* Why did things go well or not well?
        - *Conclusion:* What did you learn?
        - *Action Plan:* What would you do differently next time?

Submission Guidelines

In the LearnJCU Assignment 1 app submission dropbox, upload the following:

- A .zip file containing your Android project, exported from Android Studio (*File* > *Export* > *Export to zip file...*).
- A .pdf file containing your 500-word self-reflection.
- Provide a link to your GitHub repository in your submission. Ensure that it is shared with your lecturer and subject coordinator.

Academic Integrity

- This is an **individual assignment**. You are expected to complete this assignment independently. Proper citations should be provided for any sources or content that is not original. Any form of academic misconduct, including plagiarism, collusion, and inappropriate use of GenAI, will be pursued according to the [Coursework Academic Integrity Policy and Procedures]().
- Any use of generative AI must be appropriately acknowledged and include a [Declaration of AI-Generated Material]() (submit this with the assignment).

Marking Rubric

<table>
<tr><th>Criterion</th><th>Weight</th><th>Excellent
(85-100%)</th><th>Good
(65-84%)</th><th>Satisfactory
(50-64%)</th><th>Limited
(25-49%)</th><th>Very Limited
(0-24%)</th></tr>
<tr><td>General Code Quality</td><td>20%</td><td>The source code is very well constructed having effective code formatting, readability and naming conventions. Code comments are useful and clear.</td><td rowspan="6">Exhibits aspects of excellent (left) and satisfactory (right).</td><td>Some aspects of the source code could be formatted better or have better naming conventions. Some comments aren’t useful or clear.</td><td rowspan="6">Exhibits aspects of satisfactory (left) and very limited (right).</td><td>The source code is very poorly formatted, hard to read, and hard to understand.</td></tr>
<tr><td>App Purpose</td><td>20%</td><td>The app’s purpose is very clearly a “Utility app”. The app UX suits this purpose very well. The app provides effective “at a glance” information. The app focuses on the most important and/or relevant information.</td><td>The app’s purpose as a “Utility app” is reasonable but not outstanding. The app UX presents some useful information “at a glance”. The app focuses on some important and/or relevant information.</td><td>The app’s purpose is not reasonable. The app doesn’t present information “at a glance”. The app doesn’t provide important or relevant information.</td></tr>
<tr><td>Android API Usage</td><td>20%</td><td>The most appropriate modern APIs are used to implement the app. The use of these APIs is insightful / creative.</td><td>Some appropriate modern APIs are used to a basic level.</td><td>Poor use of available modern APIs.</td></tr>
<tr><td>Jetpack Compose Usage</td><td>20%</td><td>The app shows very effective use of Jetpack Compose to develop modular user interfaces.</td><td>Some aspects of Jetpack Compose usage could be improved.</td><td>Poor use of Jetpack Compose.</td></tr>
<tr><td>GitHub Usage</td><td>10%</td><td>A GitHub repository is used that is well-organised with clear, frequent commits that demonstrate continuous progress. Excellent README documentation.</td><td>GitHub repository is used and is functional but commits may be infrequent. The README may be missing some key details.</td><td>The GitHub repository is not used or is disorganised, with little to no commits and no README file.</td></tr>
<tr><td>Self-Reflection</td><td>10%</td><td>The self-reflection is insightful and demonstrates a detailed understanding of Android coding so far and describes meaningful experiences that occurred while working on the Utility app.</td><td>The self-reflection demonstrates some understanding of Android coding so far and describes basic experiences that occurred while working on the Utility app. The descriptions could be improved.</td><td>The self-reflection is very superficial or missing. Demonstrates very little understanding of Android coding so far and poorly describes experiences that occurred while working on the Utility app.</td></tr>
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